Improving Detection and Segmentation Performance of Prohibited Items Detection Task Models

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Xu et al. [1] propose a weakly supervised attention-based CNN for prohibited item detection in airport X-ray images, trained on an augmented GDXray dataset (5,000 images) using transfer learning with GoogLeNet; localization is achieved via high-level semantic feedback (deconvolution-based attention maps) refined by lateral inhibition and contrastive top-down attention to suppress clutter interference. The model achieves 97.6% classification accuracy and 56.6% IoU for localization (vs. 34.3% for traditional deconvolution), with per-class recognition up to 99.2% and average positioning accuracy of 65%, processing images in 0.76 seconds. However, evaluation is limited to GDXray, localization remains coarse, positioning partly relies on subjective human judgment, and the framework functions as a classifier with post hoc attention rather than an end-to-end detector.

Pai et al. [2] evaluate Faster R-CNN, STN-Faster R-CNN, and RFBNet for prohibited item detection under heavy occlusion using a large annotated subset of SIXray, introducing a tailored preprocessing pipeline and a graph-based Occlusion Index to measure overlap severity. Adding a Spatial Transformer Network (STN) before Faster R-CNN improves mAP (0.4285 vs. 0.4127) and recall (0.5467 vs. 0.5234), with about 35-40% of transformations meaningfully correcting spatial misalignment, while RFBNet underperforms. However, the affine STN applies only global warping, struggles in complex multi-object overlap, and the work is positioned as a proof-of-concept rather than a deployment-ready system.

Park et al. [3] propose a real-time YOLOv5-based detector for X-ray security screening, trained on 6000 annotated suitcase images covering 22 prohibited item classes and different overlap stages. The model achieves mAP@50 of 0.87, F1-score of 0.86, and runs at roughly 12–14 ms per image (72 FPS), making it suitable for airport deployment. However, performance drops as occlusion increases (from 0.942 to 0.762 mAP@50 across overlap stages), showing that heavy object overlap remains a major challenge.

Bajaj and Pandey [4] use an SSD300 detector with a ResNet50 backbone, applying transfer learning from COCO and fine-tuning on 200 X-ray images from the GDXray dataset to detect harmful items in baggage. The model performs reasonably well on small objects such as knives, scissors, and blades, with reported accuracies around 72–75%, but struggles to detect guns and multiple objects within the same image.

Wu et al. [5] survey deep learning methods for X-ray baggage detection, covering CNN-based, Transformer-based, and hybrid models along with key datasets such as GDXray, SIXray, and PIDray. They report that CNN-based detectors (e.g., YOLO, Faster R-CNN) generally perform better on X-ray images than pure Transformer models, especially under heavy object overlap. The review highlights persistent challenges, including severe occlusion, dataset limitations, and the need for hybrid models that better combine texture and contour features.

Mery et al. [6] present GDXray, an early public dataset for X-ray image analysis containing over 19,400 images across categories such as castings, welds, and baggage. The baggage subset includes annotated examples of guns, knives, and razor blades under realistic clutter and overlap, making it a foundational benchmark for X-ray security research. The paper does not introduce a detection model, but instead contributes a standardized dataset for algorithm development and evaluation.

Zhang et al. [7] introduce PIDray, a large-scale X-ray benchmark for real-world prohibited item detection, containing 124,486 images across 12 categories with both bounding box and instance-level mask annotations. The dataset emphasizes deliberately hidden and heavily occluded items and provides test splits (easy, hard, hidden) to reflect practical inspection difficulty. The authors also propose a divide-and-conquer, tree-structured pipeline to address the long-tailed distribution by separating binary screening from fine-grained detection and segmentation. While the benchmark significantly advances realism and scale, the study notes remaining challenges such as object orientation handling and the need for further dataset expansion.

Miao et al. [8] introduce SIXray, a large-scale X-ray benchmark for prohibited item discovery in overlapping baggage images, containing over 1 million scans with 8,929 annotated instances across six classes. The dataset reflects real-world challenges such as severe object overlap and extreme class imbalance, where positive samples account for less than 1% of the data. To address this, the authors propose Class-Balanced Hierarchical Refinement (CHR), which leverages top-down feature refinement and a class-balanced loss to improve recognition under imbalance. While CHR improves performance, the method relies on a simplified penetration-based modeling assumption, suggesting the need for more accurate physical modeling of X-ray overlap effects.

Wei et al. [9] address occluded prohibited item detection in X-ray security images by introducing the OPIXray benchmark, a cutter-focused dataset with 8,885 annotated images and test splits organized by occlusion level. To handle heavy overlap and material mixing, they propose the De-occlusion Attention Module (DOAM), a plug-and-play component that combines edge guidance and material-aware feature aggregation to refine detector features. Integrated into detectors such as SSD, YOLOv3, and FCOS, DOAM consistently improves mAP, particularly under severe occlusion. However, the gains are incremental and the approach remains an enhancement module rather than a fundamentally new detection framework, with performance still sensitive to extreme overlap conditions.

Tao et al. [10] present HiXray, a large-scale real-world X-ray benchmark with 102,928 annotated instances across 8 prohibited item categories collected from airport inspections. They propose the Lateral Inhibition Module (LIM), which suppresses noisy neighboring responses and enhances object boundaries through bidirectional propagation and boundary activation. Integrated into detectors such as SSD, FCOS, and YOLOv5, LIM improves detection performance, especially under severe occlusion. However, it functions as a plug-in enhancement rather than a fundamentally new detection framework and is mainly validated in airport security scenarios.

Ma et al. [11] introduce PIXray, the first public instance-level X-ray segmentation dataset (5,046 images, 15 categories), and propose the Dense De-Overlap Attention Snake (DDoAS) with modules to suppress background overlap and refine object contours. The approach effectively handles X-ray–specific challenges such as overlapping objects and large shape/scale variation, but remains less accurate under extreme overlap, making it a strong dataset- and method-level reference for X-ray prohibited item detection.

Zhao et al. [12] target overlapped prohibited-item detection in X-ray

security images, where objects blend with similar backgrounds or overlap with other items, making visual cues unreliable. They contribute a new CLCXray dataset (cutters + liquid containers) and proposes a Label-aware (LA) mechanism that uses label/gradient-guided feature reweighting to better separate overlapped objects, validated on CLCXray and OPIXray. It handles X-ray overlap cases (object–background similarity and multi-object overlap), but the main limitation is modest gains and reliance on label/pseudo-label guidance rather than a fundamentally new detector.

Tao et al. [13] Tao et al. present LDXray, a large-scale dual-view X-ray dataset with 146,997 paired images and 353,646 instances across 12 categories, designed to address severe occlusion in single-view detection. They propose AENet, which fuses main-view detection with an auxiliary-view pipeline using saliency localization and expert models for hard categories. The dual-view mechanism significantly improves challenging classes (e.g., +24.7% AP for umbrellas), but relies on indirect supervision for the auxiliary view due to the absence of ground-truth annotations.

Liang et al. [14] enhance YOLOv4 for prohibited item detection in X-ray images by integrating the lightweight CBAM attention module into both the backbone and detection heads, aiming to better focus on informative channels and spatial regions. Using the SIXray dataset (five categories), they report an mAP improvement from 83.39% (transfer-learning YOLOv4) to 85.51% with CBAM, with notable AP gains for knives (+4.5%) and wrenches (+4.79%). Although the added attention introduces negligible parameter overhead, the model still struggles with heavily overlapped and small objects, making the contribution a modest architectural refinement rather than a fundamental solution to X-ray overlap challenges.

Yuan et al. [15] enhance YOLOv7 for X-ray prohibited item detection by adding a Large Kernel–based Spatial Attention (SA) module to better capture long-range and contextual features under cluttered conditions. Tested on the SIXray dataset (8,908 images, five classes), the improved YOLOv7-SA raises mAP from 86.55% to 87.34%, with only a slight increase in parameters and a small drop in speed (80.1 to 77.5 fps). While it improves detection of overlapping and closely packed items, the accuracy gain is relatively small, making it an incremental architectural refinement rather than a major breakthrough.

Li and Musha [16] enhance YOLOv7 for X-ray prohibited item detection by adding Bottleneck Transformers with Hydra Attention and integrating convolution to capture both global and local features. On the SIXray dataset, their model improves mAP@0.5 from 90.6% to 95.2%, outperforming several YOLO and SSD baselines. However, evaluation on limited extreme-overlap samples and increased model size reduce its practicality for lightweight deployment.

Liu et al. [17] introduce X-Adv, a physical-world adversarial attack specifically designed for X-ray prohibited item detectors. Instead of manipulating image textures (which are largely suppressed in X-ray scans), the method generates small 3D-printable metal objects with optimized shapes that can mislead detectors when placed inside luggage. By modeling X-ray attenuation and using reinforcement learning to find effective placement locations, the attack significantly reduces detection performance—for example, lowering DOAM's mAP from 74.02% to 23.05% in experiments. Although the attack is highly effective, its impact decreases under random placement and can be partially reduced through adversarial training, showing both the seriousness of the vulnerability and the possibility of defense.

Liang et al. [18] propose a lightweight model built on YOLOv4-tiny, designed for efficient X-ray prohibited item detection under clutter and overlap using depthwise separable convolutions, SE attention, and a dilated convolution spatial pyramid for efficient multi-scale feature extraction. It is trainied on a self-collected and augmented dataset of 5015 images and 8 classes, using techniques such as flipping, blurring, affine, and Mosaic augmentation to simulate real scenarios. The model achieves strong performance, reaching 95.59% mAP at 122 FPS. However, performance is still affected by complex backgrounds and occlusion, and reliance on a custom dataset limits generalization.

Zhang et al. [19] explore a weakly supervised prohibited item detection and segmentation in X-ray images to reduce labeling cost under clutter and overlap. They proposes an attention-based pipeline refined with inhibition and mask generation through SLIC/GrabCut, evaluated on SIXray and a self-built dataset contains16k images. The method addresses X-ray challenges like overlap, small objects, and visual similarity, achieving 54.26% localization IoU and 65.78% segmentation IoU on SIXray. However, the method is slower, depends on classifier quality, and struggles under heavy overlap, highlighting the need for larger datasets and more robust models in future work.

Li et al. [20] propose a Dual Attention Mechanism Network (DAMN) combining spatial and channel attention with dependency refinement to capture both local long-range feature relationships. Built on a ResNet-101-FPN backbone and evaluated on PIDray dataset. The model improves detection and segmentation performance, achieving 63.8% detection AP and 54.7% segmentation AP, with strong gains on hidden (occluded) cases. The method effectively handles overlap, small objects, and detail loss by enhancing feature representation. However, it is computationally heavier than one-stage models and still slightly underperforms YOLO-based methods on some easier cases, indicating a trade-off between accuracy and efficiency.

Tao et al. [21] introduce the first X-ray FSOD dataset (12,333 images, 41,704 instances, 20 classes) and proposes the Weak-feature Enhancement Network (WEN), which boosts detection by aggregating and refining weak features using prototype-based learning. They address few-shot prohibited item detection in X-ray images, where rare classes and weak visual features (due to occlusion and color fading) limit performance. Built on a Faster R-CNN framework, WEN significantly improves mAP over existing FSOD methods, especially in low-shot settings. Its main limitation is reliance on a two-stage fine-tuning process, which may limit scalability to larger or real-time settings.

Chen et al. [22] propose ZPGNet, an atomic-number prior guided network for prohibited item detection in cluttered X-ray images, take advantage of material information derived from X-ray color encoding to remove background interference. Using three modules: ZPG for creating atomic-number maps, MA for combining material and image features, and BE for improving multi-scale representation—the method extracts material cues from X-ray colors rather than depending solely on visual features. Tested on HiXray and OPIXray, it handles X-ray challenges like penetration overlap and low texture, achieving 83.9% mAP50 and 90.7% mAP50, outperforming DOAM/LIM. However, the approach depends on accurate material estimation and still struggles with severe occlusion, which remains an open challenge.

Zhang et al. [23] propose TinyRay, a lightweight detection model based on an improved YOLOv7-tiny, designed for fast and efficient X-ray prohibited item detection on limited hardware. The model replaced the FasterNet backbone, introduces a PConv-based New-ELAN module to reduce computation, and adds a coordinate attention mechanism to enhance small-object detection. Evaluated on the SIXray dataset, the model achieves about 90.2% mAP@0.5 with faster inference (89 FPS) while using fewer parameters than the original YOLOv7-tiny. Nevertheless, under severe occlusion and clutter, performance still declines, and the enhancements primarily concentrate on efficiency rather than providing a fundamental solution to X-ray-specific problems.

Lin et al. [24] introduces OVXD, which adapts CLIP to X-ray images using a lightweight feature adapter, tested on PIXray and PIDray. The method handles domain shift, overlap, and weak textures, achieving 21.0 AP50 on PIXray and 27.8 AP50 on PIDray, outperforming prior open-vocabulary baselines. Its limitation is that novel-class accuracy remains low, so it's best cited for zero-shot/open-set X-ray detection rather than standard detection.

Cani et al. [25] present a comprehensive benchmarking study for X-ray prohibited item detection, addressing inconsistencies in prior evaluations. It compares CNN, transformer, and hybrid detectors across six public X-ray datasets (including SIXray, OPIXray, HiXray, PIDray) under unified metrics. The analysis highlights key X-ray challenges such as overlap, clutter, low texture, and scanner domain shift, showing that no single model works best in all cases. CNNs are fastest, transformers handle domain shift better, and hybrids often perform best under heavy occlusion.

Cani et al. [26] study hybrid CNN–transformer models for X-ray

prohibited item detection, focusing on whether they improve robustness over standard CNNs. They compare YOLOv8 with hybrid variants using Next-ViT-S and RT-DETR on EDS, HiXray, and PIDray. The study shows that YOLOv8 (CNN-only) performs best on HiXray (mAP50 = 0.845) and PIDray (0.897), while hybrid models are more robust under domain shift, achieving the best results on EDS (mAP50 0.588). The study primarily functions as a comparative analysis of hybrid architectures for enhancing robustness, highlighting important issues like occlusion, clutter, and scanner variability.

Gan et al. [27] propose YOLO-CID, an improved YOLOv7-based model for X-ray contraband detection to better handle small, low-contrast, and occluded objects in cluttered bags. The method introduces an MP-OD module for better feature localization, a BiFPN-P3 structure for improved multi-scale fusion, and a shuffle attention mechanism to focus on important features. Experiments on PIDray and CLCXray show clear gains, reaching 70.3% AP50 (PIDray) and 80.2% AP50 (CLCXray) at 40 FPS. The main limitation is that it is an incremental YOLO improvement, making it suitable to cite as a YOLO-based enhancement, not a new detection framework.

Ren et al. [28] address X-ray prohibited item segmentation under heavy overlap and occlusion, where natural-image models fail. They integrates SAM with Mask R-CNN using a bilayer occlusion-aware decoder and introduces PIDray-A and PIXray-A with added occlusion annotations. Tested on these datasets, it achieves mAP 57.6 (PIDray-A) and 58.9 (PIXray-A), slightly outperforming strong baselines. The main limitation is the need for extra occlusion labeling and limited gains on hidden items, so it's best cited as an occlusion-aware segmentation extension rather than a general detector.

Xiang et al. [29] propose a material-aware path aggregation network for X-ray prohibited item detection, built on YOLOX to better handle multi-scale objects and occlusion. The method introduces a smoothed atrous convolution (SAC) module for multi-scale feature extraction, a material-aware coordinate attention (MCA) mechanism to capture material cues, and a shape-decoupled SIoU loss to improve localization under varying aspect ratios. Experiments on OPIXray and SIXray show strong performance, achieving up to 92.65% mAP and outperforming several state-of-the-art methods. However, the approach increases computational cost and mainly relies on architectural enhancements, making it an effective but incremental improvement over existing YOLO-based detectors.

Gao et al. [30] propose an improved YOLOv8s-based model for X-ray contraband detection, enhancing feature extraction and robustness under occlusion and complex backgrounds. The approach combines an EMA attention mechanism to reduce noise and overlap, SRGAN for data enhancement, DCNv2 for adaptive feature extraction, and an enhanced pigeon-inspired optimization to adjust the learning rate. Improved performance is demonstrated by experiments on a self-built dataset, which reach approximately 73.4% mAP with real-time speed (95 FPS). However, the approach limits comparability and generalization to standard benchmarks because it primarily introduces incremental improvements and depends on a private dataset.

Velayudhan et al. [31] study baggage threat recognition in X-ray images under severe class imbalance and clutter, where CNNs struggle with global context. It is one of the first to apply Vision Transformers with focal loss, combined with weakly supervised localization, to handle low texture, occlusion, and long-tailed positives. Evaluated on Compass-XP and SIXray, it achieves strong results (F1 = 0.979 and 0.873, respectively). Its main limitation is that it focuses on image-level classification, not full object detection or segmentation, making it best cited for imbalanced X-ray threat recognition rather than detection.

Wang et al. [32] propose a lightweight YOLOv8-based approach for X-ray prohibited item detection under occlusion, clutter, and scale imbalance. The method ASFF, coordinate attention, slide loss, and Soft-NMS to better handle overlapping and hard samples, evaluated on PIDray and SIXray. The method achieves 90.2/90.5/79.1 mAP50 on PIDray Easy/Hard/Hidden and 91.4 mAP on SIXray with only 3M parameters. However, performance remains weaker for certain categories, degrades under severe occlusion, and lacks analysis of false alarms in real-world deployment.

Zhang et al. [33] build a balanced X-ray dataset (13,728 images, 12 classes) by combining SIXray and PIDray to address class imbalance, and propose two YOLOv11s-based models, ASEA-Net and CSEC-Net. While CSEC-Net reduces parameters (-39.91%) and FLOPs (-40.38%) for effective deployment, ASEA-Net achieves 95.78% accuracy and 93.55% mAP@50 with reduced parameters (-13.37%). Although the models enhance detection performance in challenging situations like clutter and occlusion, generalization to real-world distributions may be constrained by their dependence on a carefully constructed balanced dataset and small architectural changes.

Teng et al. [34] propose STRay, an improved YOLOv7-based model for X-ray prohibited item detection to address occlusion and small-object challenges. The technique incorporates an effective decoupled head to enhance localization and classification, a Swin Transformer to reduce background noise, and deformable dilated convolutions to better capture object shapes under overlap. It outperforms several baseline detectors, achieving 95.3%, 88.8%, and 83.1% mAP on SIXray, OPIXray, and PIDray. The future work will focus on improving the model's ability to detect multi-scale contraband items and further enhance detection accuracy.

Sun et al. [35] propose DV-DINO, a dual-view X-ray prohibited item detection model built on the DINO transformer to address occlusion and viewpoint limitations in single-view systems. In order to combine complementary features based on shared spatial information, the method introduces a VDIS module and processes vertical and horizontal views concurrently. When tested on the Dualray dataset, DV-DINO outperforms baseline DINO in detection performance by +2.1% mAP and demonstrates superior robustness in the face of occlusion and distorted imaging. The outcomes show how well the suggested dual-view framework works to increase detection accuracy in challenging X-ray security situations.

Shao et al. [36] propose an FBS (foreground–background separation) framework for X-ray prohibited item detection to address severe object overlap and background interference. The technique applies attention to focus detection on pertinent regions after reconstructing foreground objects using recursive training and a decoder guided by X-ray imaging properties. It consistently outperforms YOLOv4, DOAM, and CHR in mAP and recall when tested on synthetic and real datasets (GDTIPXray, OPIXray, and SIXray). Future research on better separation and weakly supervised learning is suggested by the fact that performance gains are restricted when forbidden items overlap with one another and that accurate foreground reconstruction without ground truth remains difficult.

Xiong et al. [37] propose DoubleRYOLO, an extension of YOLOv4 for rotated prohibited item detection in X-ray images to better handle objects with varying orientations and shapes. With a modulated rotation loss to enhance angle regression, the model employs two prediction branches: one for oriented bounding boxes and another for horizontal bounding boxes. In comparison to SSD, YOLOv4, and other baselines, it achieves higher recall (up to 77.65%) on a large-scale Knife-DH dataset (664k images), indicating that oriented bounding boxes provide more accurate localization. However, the dual-branch design adds extra complexity and concentrates primarily on knife detection.

Nguyen et al. [38] propose a real-time X-ray prohibited item detection framework that combines adaptive multiscale convolution blocks (AMC) with a lightweight dynamic convolution module (ALCM) to improve feature extraction across different object scales. Robust detection under clutter and occlusion is made possible by the model's dynamic fusion of both fine-scale and large-scale features using adaptive weighting. It maintains real-time speed (42 FPS) while achieving 91.2% mAP@0.5 on OPIXray and 87.3% mAP@0.5 on HiXray. However, generalization across various scanners is still difficult, and performance deteriorates in highly cluttered scenes, indicating future work on domain adaptation and enhanced contextual modeling.

Ren et al. [39] propose LightRay, a lightweight YOLOv4-based model for X-ray prohibited item detection that replaces the backbone with MobileNetV3 and introduces a feature enhancement module combining LFPN and CBAM to improve multi-scale feature fusion . The model's main goal is to enhance small-object detection in X-ray images that are obscured and cluttered. When tested on the SIXray dataset, it achieves

87.28% mAP while maintaining real-time speed (42 FPS) and drastically lowering computational cost (about 1/5 FLOPs and 1/3 parameters). The model's limitations in complex scenarios are highlighted by the fact that it still exhibits lower recall for small and heavily occluded items.

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